Lightweight hallucination span detection

Source-conditioned token classification with exact character-span output

a Inference architecture

Ensemble configurations are candidates for evaluation

SOURCE PROMPT
The trial enrolled 120 adults.

LLM ANSWER
The trial enrolled 150 adults.

Illustrative input

Pair tokenizer
Prompt + answer
4,096-token context
Answer offsets [start, end)

Truncate prompt only

INDEPENDENT MODEL WEIGHTS

Hard-target models
ModernBERT-base · 0.15B each
Seeds 42 / 43 / 44

Soft-target models
ModernBERT-base · 0.15B each
Seeds 42 / 43 / 44

Published LettuceDetect-large
Optional member · 0.4B

Token probabilities
Two-class softmax

Equal-weight mean
$\bar{p}_i = (1/M)\ \Sigma_m\ p_i^{(m)}$

Same tokenizer and identical ordered offsets

Span decoder
1 Probability threshold
2 Merge token gaps
3 Optional word snap

Per-task calibration
See protocol in panel c

CHARACTER-SPAN OUTPUT
The trial enrolled **150** adults.

[19, 22) → "150"
Unsupported answer characters

Zero-based, end-exclusive offsets

b Training supervision

Annotated character spans supervise answer tokens only.

Hard target $y_i = h_i$



Soft target $y_i = 0.6h_i + 0.4q_i$



h_i : overlap with a gold span

q_i : boundary-shift coverage weighted by $J + J^2$

J is word-level Jaccard. Target profiles above are schematic.

ANSWER-ONLY CROSS-ENTROPY (A: answer tokens)

$$L = -\ \text{mean}_{i \text{ in } A}\ [\ y_i \log p_i + (1 - y_i) \log(1 - p_i)\ ]$$

c Calibration and evaluation

Our models: split RAGTruth training data by source.

90% train sources
Fit independent classifier weights

→

10% dev sources
Select task-specific decoder settings

→

RAGTruth test
Evaluate frozen pipeline
2,700 answers

LettuceDetect exception

Any ensemble including published LD uses fixed decoding: threshold 0.5, gap 0, no word snapping.

Two-fold cross-fitting is restricted to published LD alone.

Primary metric: LettuceDetect-compatible character span F1

Paired bootstrap over test sources · 95% confidence interval

Retain a change only when its gain interval is entirely above zero.

Architecture of the current pipeline and Part C ensemble experiment. Parameter counts are per model.

Soft targets retain the legacy boundary-shift implementation. Ensemble gains await full evaluation.